Seungpil Choi

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EDUCATION

University of Minnesota Sep. 2026 – Present
Ph.D. Student in Electrical Engineering Minneapolis, MN

- Advisor: Prof. Mehmet Akçakaya

Hankuk University of Foreign Studies (HUFS) Mar. 2019 – Aug 2024
Bachelor of Computer and Electronic System Engineering Yongin, South Korea

- Advisor: Prof. Byunghwan Jeon

RESEARCH INTERESTS

- Medical Image Analysis
- Signal Processing
- Deep Reinforcement Learning

PUBLICATIONS

[Journal] Choi, Seungpil†, Seoyeon Jang†, Sunghee Jung, Heon Jae Cho, and Byunghwan Jeon. "Deep reinforcement learning for efficient registration between intraoral-scan meshes and CT images." *Pattern Recognition* 164 (2025): 111502. **(IF: 7.6)**

[Journal] Jang, Seoyeon†, Seungpil Choi†, and Byunghwan Jeon. "Reinforcement Learning-Based Centerline Tracking With Cross-Sectional Similarity for Enhancing Mandibular Canal Segmentation." *IEEE Access* (2025).

† Co-first authors

RESEARCH EXPERIENCE

Enhancing Mandibular Canal Segmentation Performance Using Reinforcement Learning-Based Centerline Extraction.
Oct. 2023 – Mar. 2025

- Developed a reinforcement learning framework for centerline tracking in 3D mandibular canal segmentation
- Designed cross-sectional similarity-based reward functions for anatomically consistent navigation

Smart Curling Coaching Systems.
Nov. 2023 – Jan. 2025

- Developed a multi-agent reinforcement learning framework for strategic curling simulation and coaching
- Implemented a physics-based curling simulator and imitation learning pipeline

Deep Reinforcement Learning for Efficient Registration between Intraoral-scan Meshes and CT Images.
Sep. 2022 – Oct. 2023

- Developed a reinforcement learning-based registration framework for 3D intraoral scan meshes and CT images
- Implemented geometric feature extraction and refinement methods for registration accuracy improvement

HONORS AND AWARDS

University of Minnesota

- Jae Moon ECE Fellowship (4 years) Fall 2026 – 2030
- CSE Fellowship (awarded to top incoming students) Fall 2026, Spring 2027

HUFS

- Scholarship for Academic Excellence Fall 2020, 2023
- Merit Scholarship Fall 2020

TEACHING EXPERIENCE

HUFS

- Teaching Assistant, Algorithm Course — Fall 2023
  - Assignment creation and grading
- Teaching Assistant, Object-Oriented Programming Course — Fall 2022
  - Laboratory Sections for Java, Assignment creation and grading

Gyeonggi-do Office of Education

- Teaching Assistant for Teacher Qualification Training Program — Jan. 2023 – Aug. 2023
  - Supported laboratory sessions for courses including Computer Programming, Algorithms, and Computer Architecture
  - Assisted participants with programming exercises in C++, Python, and microprocessor systems

SKILLS

Programming Languages : Python, Java, C/C++
Related Framework or Library : VTK, Tensorflow, Pytorch